Ensemble

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Adapted\inspired from Mahdi’s and Polo’s slides.

Which Classifier/Model to Choose?

Possible strategies:

- Go from simplest model to more complex model until you obtain desired accuracy
- Discover a new model if the existing ones do not work for you
- Combine all (simple) models

Common Strategy: **Bagging**

(**B**ootstrap **Agg**regat**ing**)

Originally designed for combining multiple models, to reduce variance and improve classification "stability" [Leo Breiman, 94]

Uses random training datasets (sampled from one dataset)

http://statistics.about.com/od/Applications/a/What-Is-Bootstrapping.htm

Common Strategy: Bagging
(Bootstrap Aggregating)

Consider the data set $S = \{(X_i, Y_i)\}_{i=1,\ldots,n}$

- Pick a sample S^* with **replacement** of size *n*
  (S^* called a "bootstrap sample")

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Common Strategy: Bagging
(Bootstrap Aggregating)

Consider the data set $S = \{(X_i, Y_i)\}_{i=1,\ldots,n}$

- Pick a sample S^* with **replacement** of size n
  (S^* called a "bootstrap sample")
- Train on S^* to get a classifier f^*
- Repeat above steps B times to get $f_1, f_2, \ldots, f_B$
- Final classifier $f(x) = \text{majority}\{f_b(x)\}_{j=1,\ldots,B}$

Common Strategy: Bagging

Why would bagging work?

- Combining multiple classifiers reduces the variance of the final classifier

When would this be useful?

- We have a classifier with high variance

Bagging decision trees

Consider the data set S

- Pick a sample S^* with replacement of size n
- Grow a decision tree T_b
- Repeat B times to get $T_1,\ldots,T_B$
- The final classifier will be

$$f(x) = \text{majority}\{f_{T_b}(x)\}_{b=1,\ldots,B}$$

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Random Forests

Almost identical to bagging decision trees,
except we introduce some randomness:

- Randomly pick *m* of the *d* available attributes, at every split when growing the tree
  (i.e., d - m attributes ignored)

Bagged **random** decision trees
= **Random forests**

Random Forests: Why does it work?

1. **Reduces variance through bootstrapping**
   Each tree is trained on a different random sample → averaging cancels noise and stabilizes predictions.
2. **Computationally cheaper than growing full trees**
   Decision trees can be expensive with many features;
   Random Forests speed this up by splitting on **only a small subset of features** at each node.
3. **Reduces correlation across trees**
   Randomly selecting a subset of features at each split forces trees to grow differently → diversity leads to a stronger ensemble.
4. **Combines many weak learners into one strong model**
   Individual trees may overfit, but averaging many overfitted trees produces a smooth, robust decision boundary.
5. **Automatically captures complex patterns**
   Learns nonlinear relationships and high-order interactions without feature engineering.
6. **Works well out-of-the-box**
   Handles noise, missing values, outliers, mixed data types, and requires little tuning.

What are our Hyper-Parameters in Random Forest

$$m = Number\ of\ randomly\ chosen\ attributes$$

Usual values for $m = \sqrt{d}, 1, 10$

d is number of dimensions or features or attributes

How to optimize m? Cross-Validation

$$B = Number\ of\ models\ or\ decision\ trees\ in\ Random\ Forest$$

Keep adding trees until training error stabilizes (reaches to a plateau)

Important points about random forests

Algorithm (hyper) parameters

- Size/#nodes of each tree
  - as in when building a decision tree
- May randomly pick an attribute, and may even randomly pick the split point!
  - Significantly simplifies implementation and increases training speed
  - PERT - Perfect Random Tree Ensembles http://www.interfacesymposia.org/I01/I2001Proceedings/ACutler/ACutler.pdf
  - Extremely randomized trees http://orbi.ulg.be/bitstream/2268/9357/1/geurts-mlj-advance.pdf